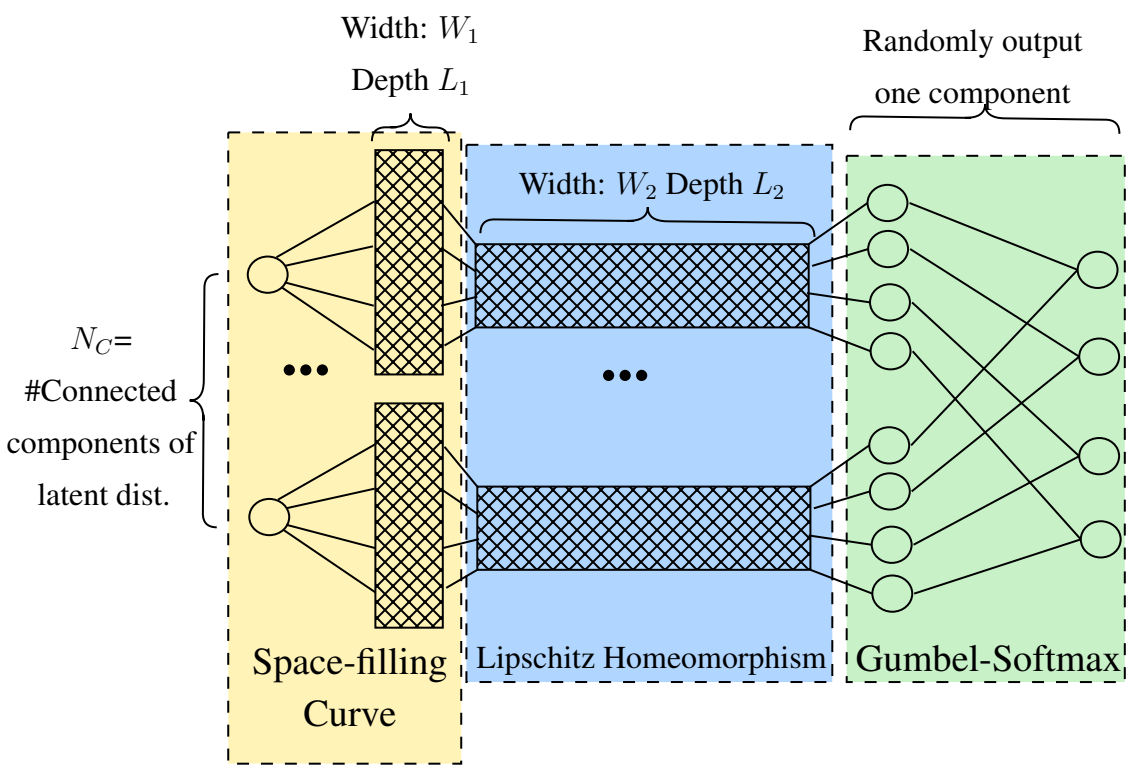
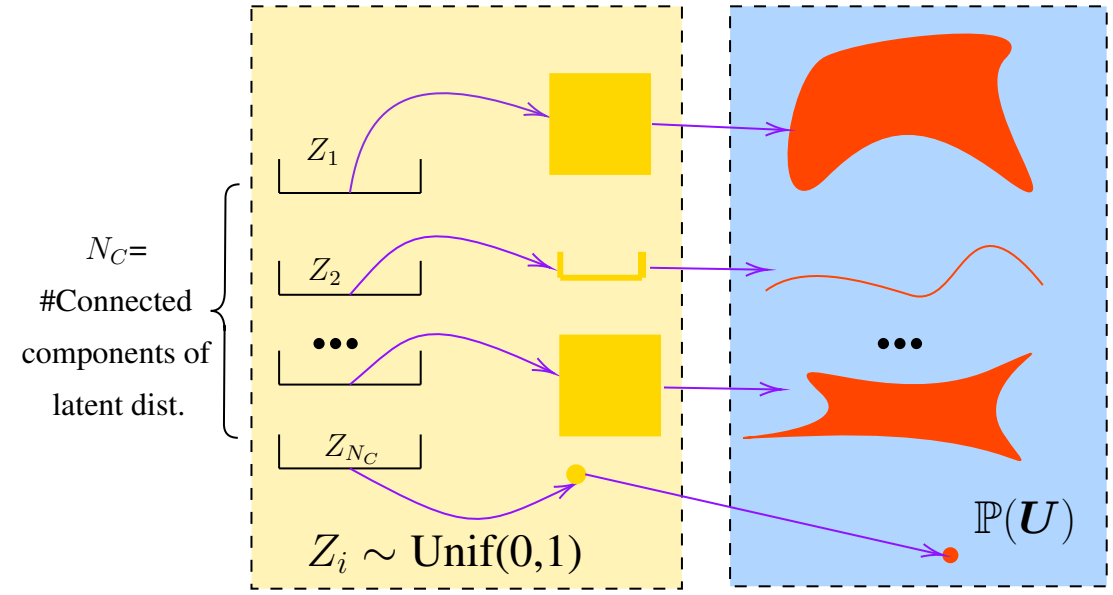
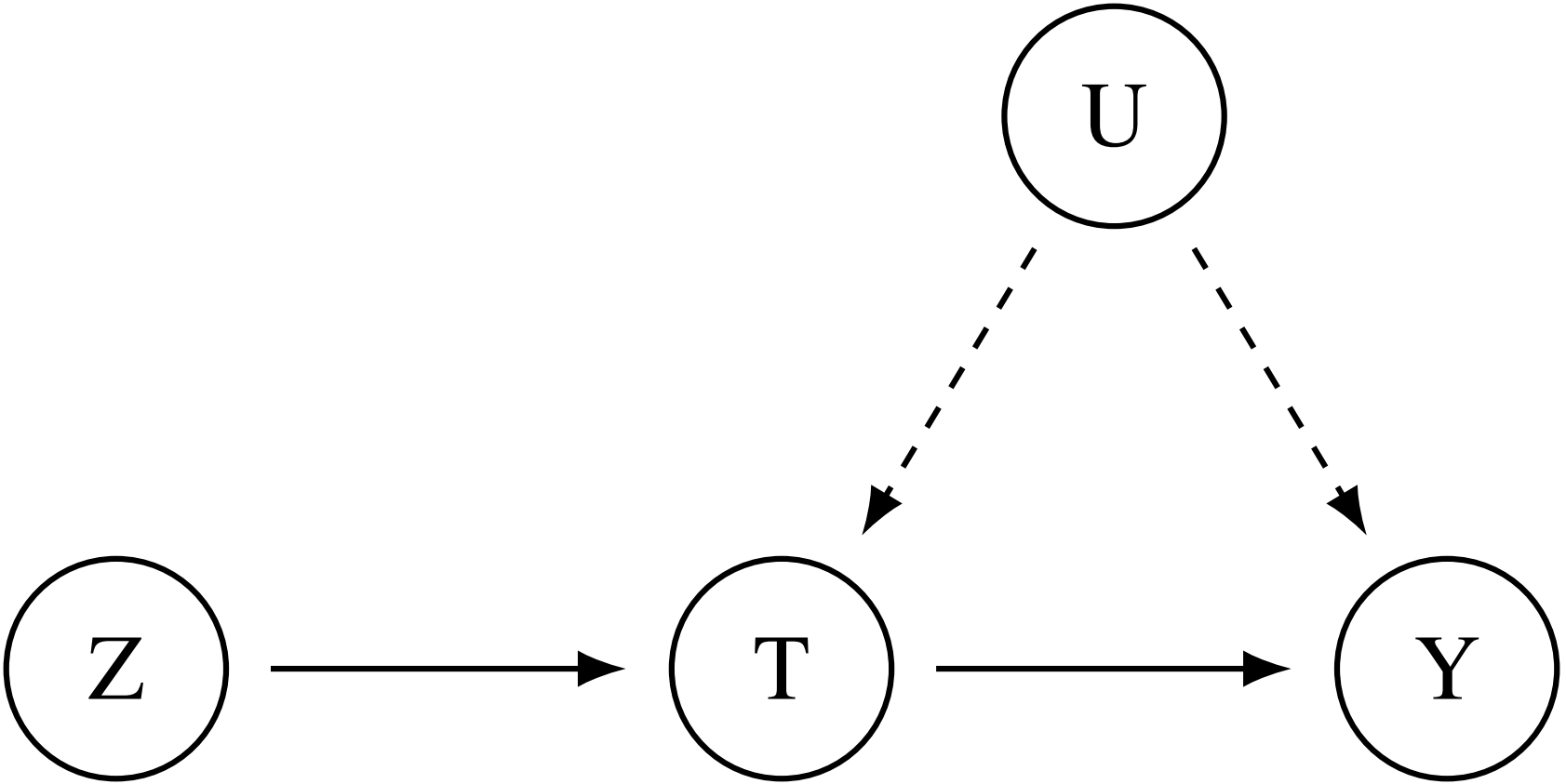
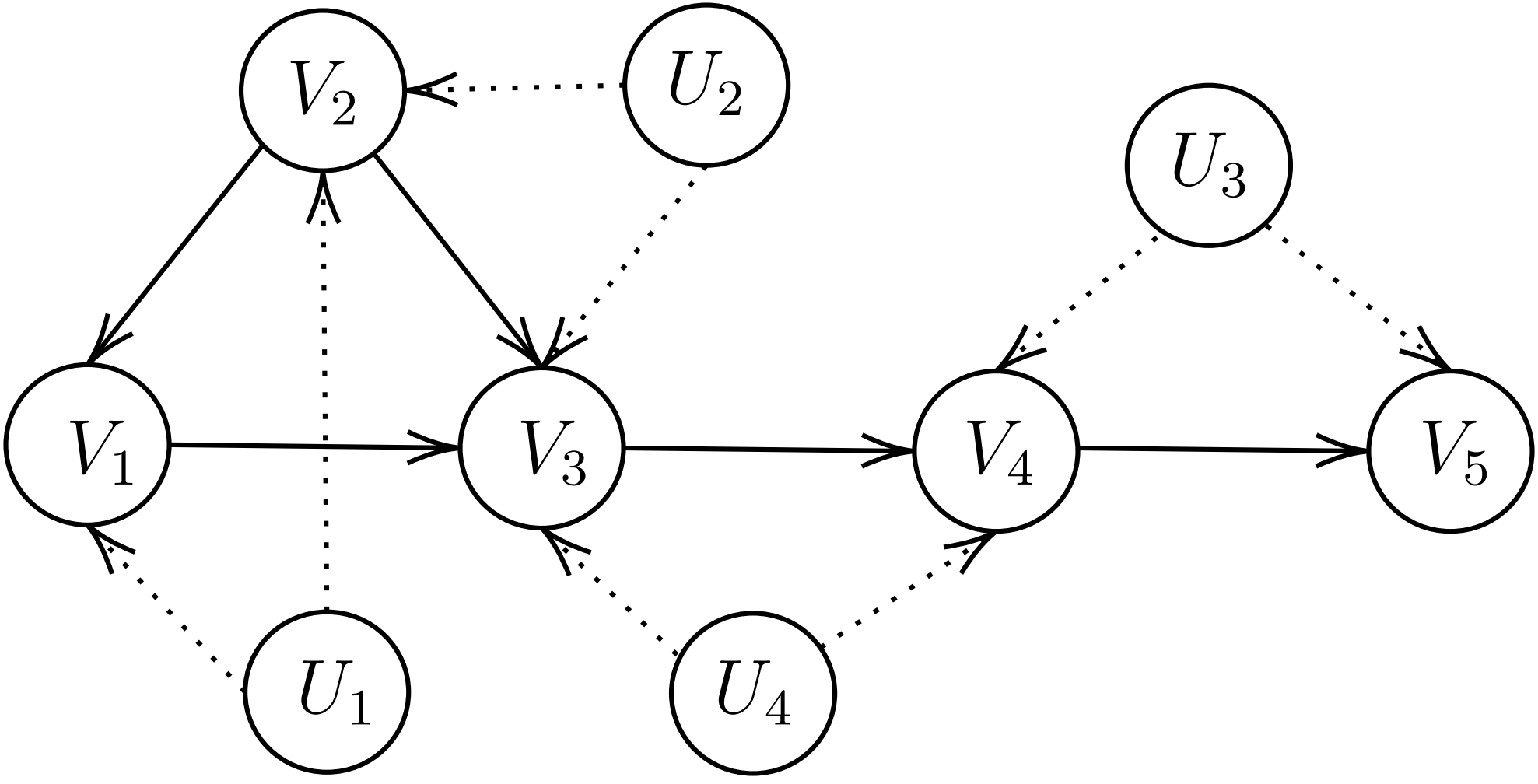
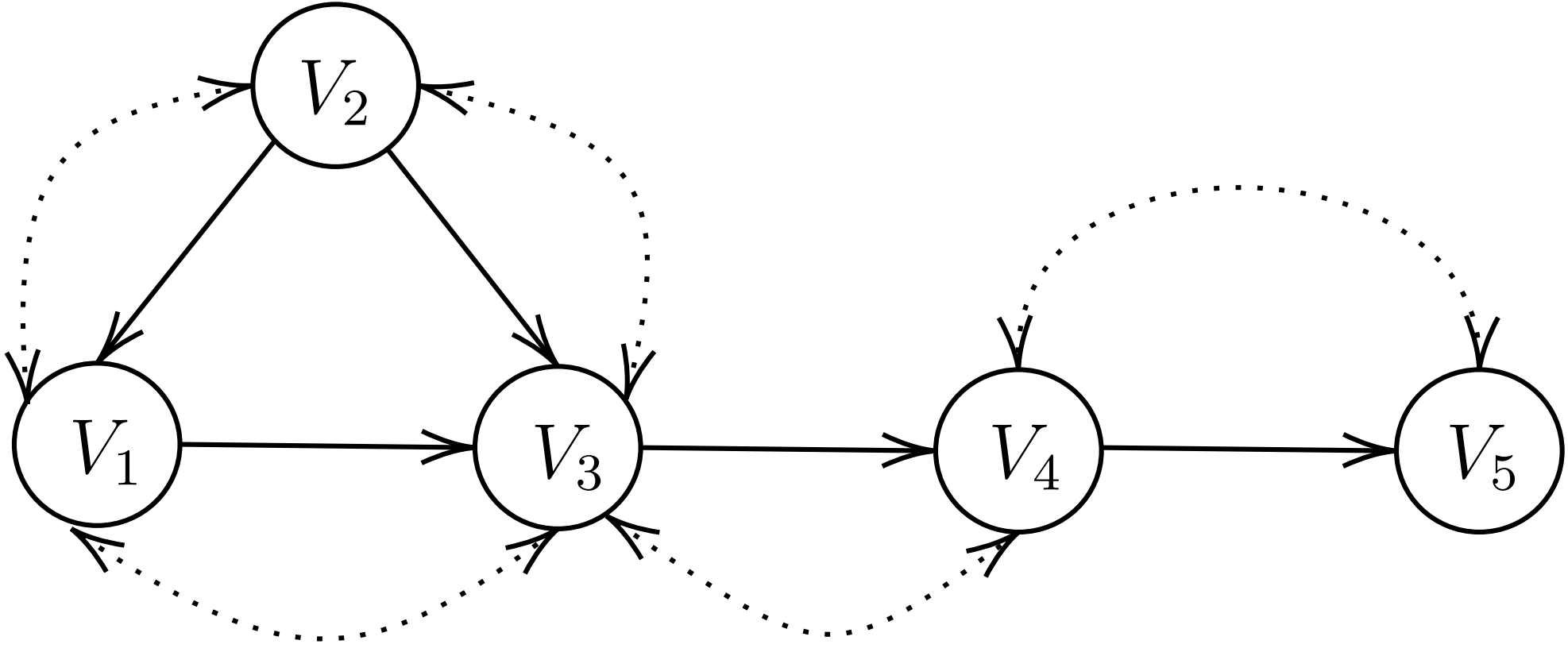
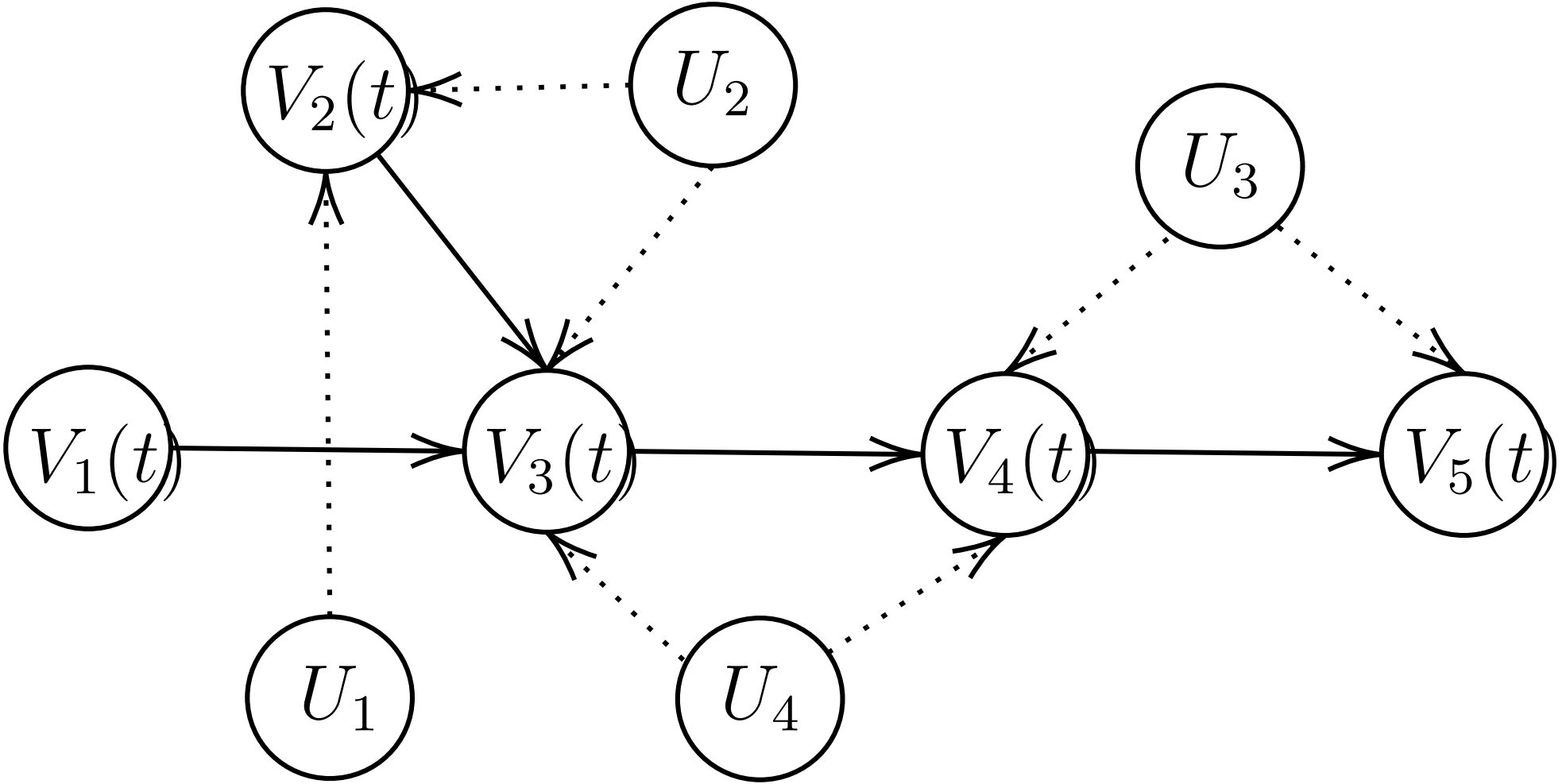

Width: W_1
Depth L_1
Randomly output
one component
Width: W_2 Depth L_2
N_C=
#Connected
components of
latent dist.
Space-filling
Curve
Lipschitz Homeomorphism
Gumbel-Softmax

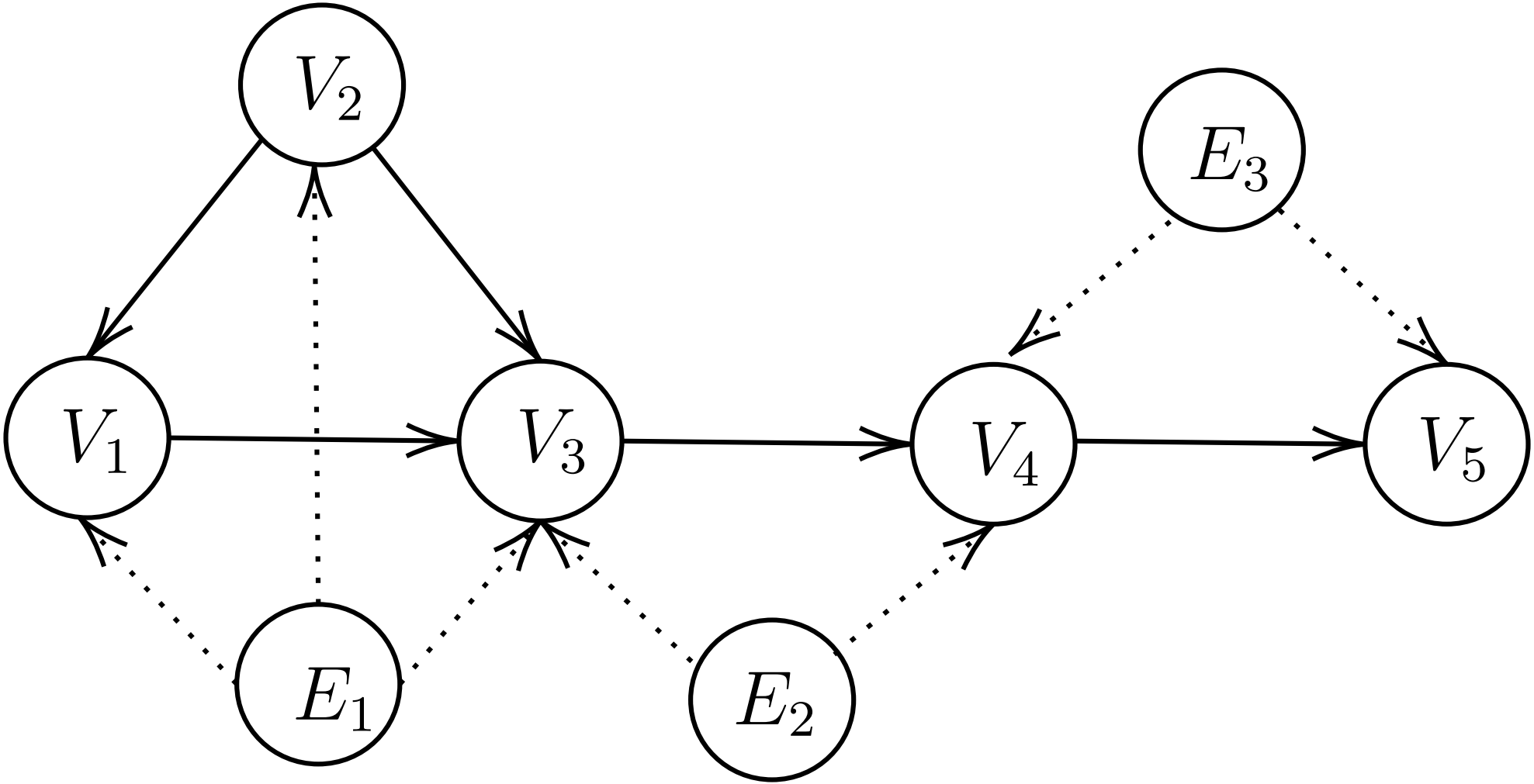

N_C=
#Connected
components of
latent dist.
Z_1
Z_2
Z_{N_C}
$Z_i \sim$ Unif(0,1)
$\mathbb{P}(\boldsymbol{U})$

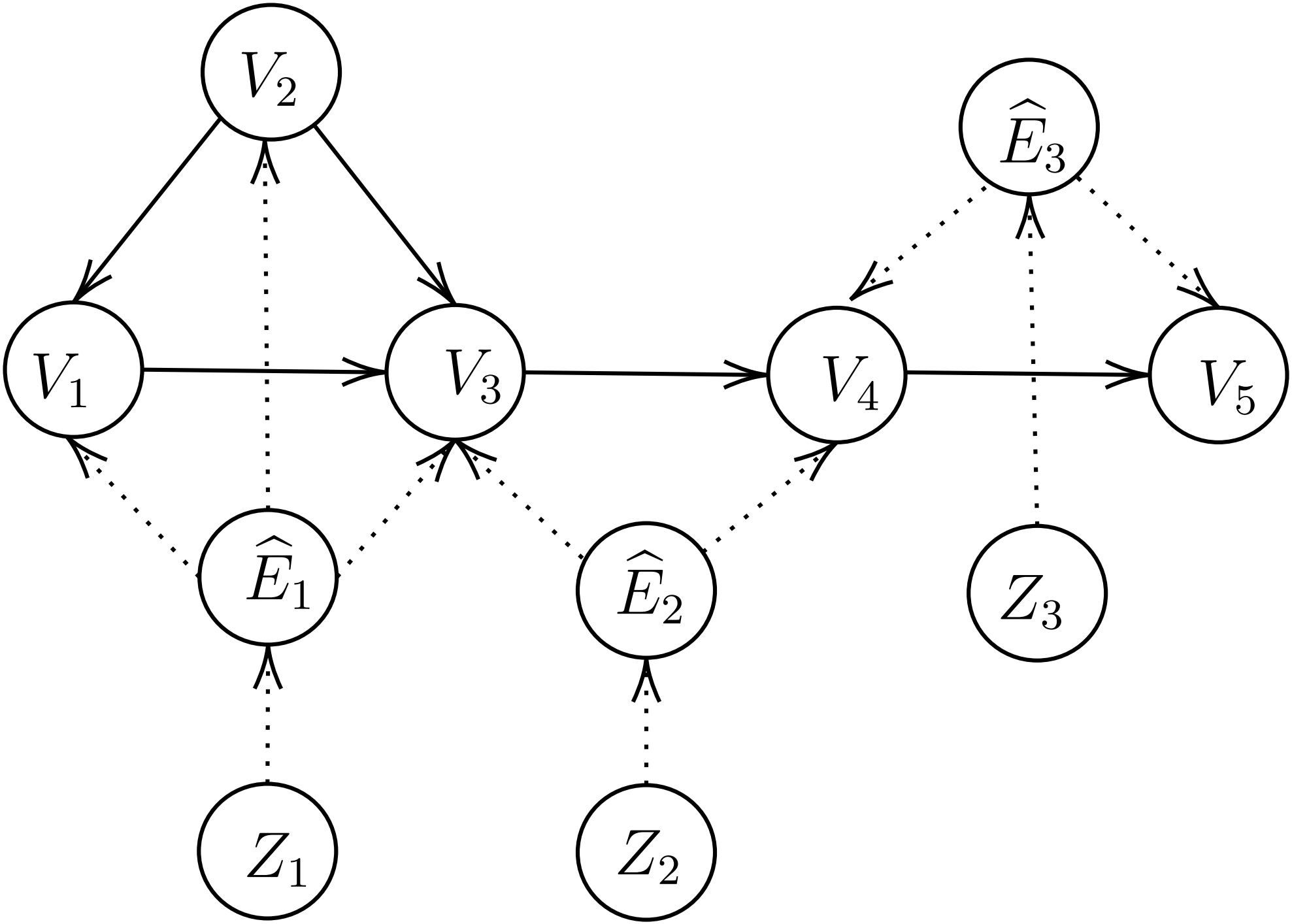

U
Z
T
Y

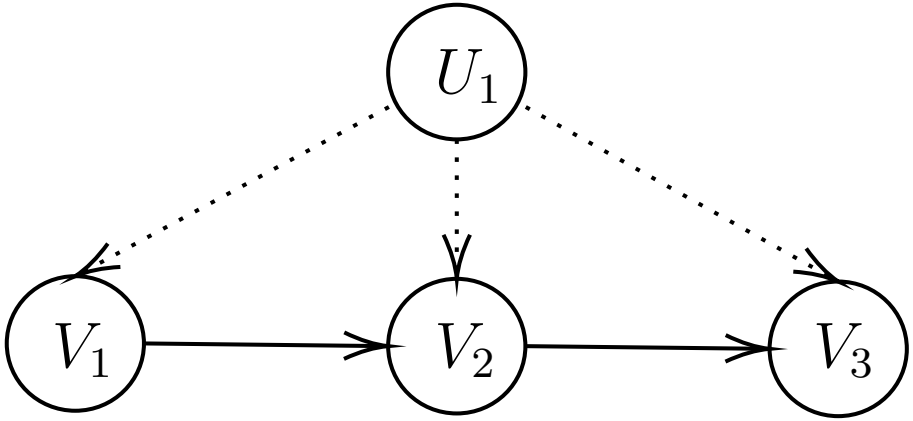

V_2
U_2
U_3
V_1
V_3
V_4
V_5
U_1
U_4

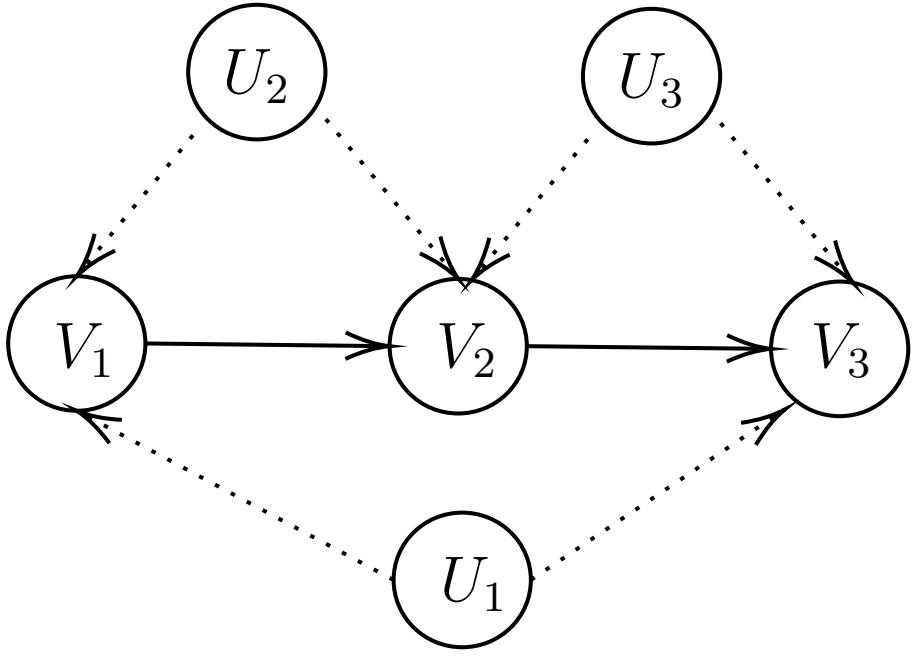

V_2
V_1
V_3
V_4
V_5

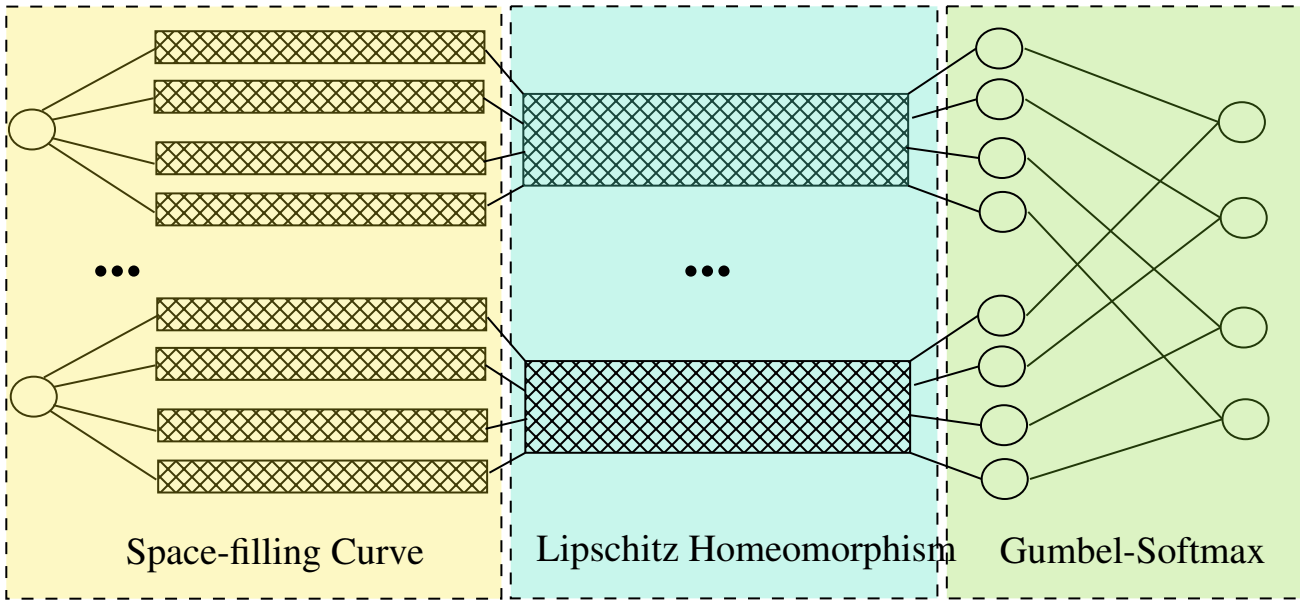

$V_2(t)$
U_2
U_3
$V_1(t)$
$V_3(t)$
$V_4(t)$
$V_5(t)$
U_1
U_4


V_2
E_3
V_1
V_3
V_4
V_5
E_1
E_2


V_2
$\widehat{E}_3$
V_1
V_3
V_4
V_5
$\widehat{E}_1$
$\widehat{E}_2$
Z_3
Z_1
Z_2


U_1
V_1
V_2
V_3


U_2
U_3
V_1
V_2
V_3
U_1


Space-filling Curve
Lipschitz Homeomorphism
Gumbel-Softmax

